Vertical, Horizontal and Oblique Asymptotes

Vertical Asymptotes (V. A.)

We can identify Vertical Asymptotes (V. A.) by finding x values that make the denominator [redacted].

$= 0$

If $x = a$ is a V. A., calculate $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$. This limit will either be $+\infty$ or $-\infty$. To decide which test values near the V. A.

Example 1: Sketch the graph of $f(x) = \dfrac{2(x^2+1)}{(x-1)(x+2)}$ near its vertical asymptotes.

V.A. @ $x = 1$

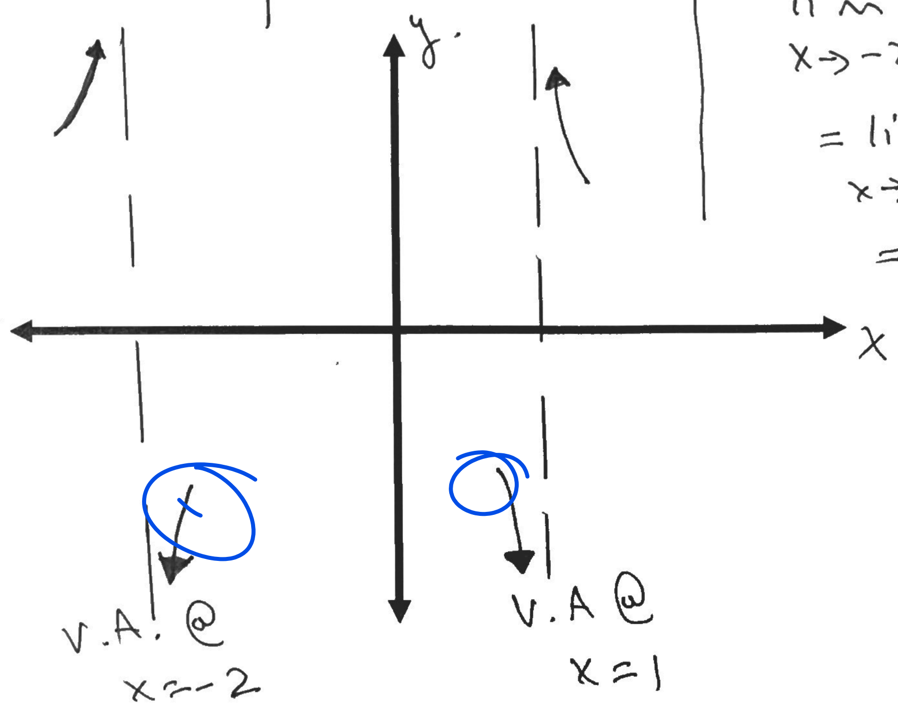
$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} \frac{2(x^2+1)}{(x-1)(x+2)} = -\infty$$

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} \frac{2(x^2+1)}{(x-1)(x+2)} = \infty$$

V.A. @ $x = -2$

$$\lim_{x \to -2^-} f(x) = \lim_{x \to -2^-} \frac{2(x^2+1)}{(x-1)(x+2)} = \infty$$

$$\lim_{x \to -2^+} f(x) = \lim_{x \to -2^+} \frac{2(x^2+1)}{(x-1)(x+2)} = -\infty.$$



Horizontal Asymptotes (H. A.)

- The location of the horizontal asymptotes can be found by determining $\lim_{x \to \infty} f(x)$ or $\lim_{x \to -\infty} f(x)$.
- Determine the value of $f(x)$ for both small and large values of x.

→ If the value is greater than the value of the H. A., then the function is approaching the H. A. from above.

→ If the value is less than the value of the H. A., then the function is approaching the H. A. from below.

Example 2: Determine the H. A. of $f(x) = \dfrac{3-2x}{x-3}$. Also determine where the function is above and below the H. A.

H. A. @ $y = -2$

* Cross test $f(x) =$ H. A.

$$\frac{3-2x}{x-3} = -2$$

$$3-2x = -2x+6$$

$$3 \neq 6$$

$\Rightarrow$ DOES NOT CROSS H. A.

$$f(100) = \frac{3-200}{100-3} = \frac{-197}{97} < -2$$

$$f(-100) = \frac{3+200}{-103} = \frac{-203}{103} > -2$$

$\therefore$ As $x \to \infty$, $f(x) \to -2^-$ ← Below

As $x \to -\infty$, $f(x) \to -2^+$ ← Above

Example 3: Determine if $f(x) = \dfrac{3x^2+2x-1}{x^2-4}$ crosses the H. A. If so, determine the point of intersection.

H. A. @ $y = 3$

$f(x) =$ H A.

$$\frac{3x^2+2x-1}{x^2-4} = 3$$

$$3x^2+2x-1 = 3x^2-12$$

$$2x = -11$$

$$x = \frac{-11}{2}$$

$\therefore$ $f(x)$ crosses the H. A. @ $\left(-\frac{11}{2}, 3\right)$

Example 4: Determine if $f(x)=\dfrac{x^3-13x+12}{x^2-3x-10}$ crosses the oblique asymptote. If so, determine the point of intersection.

$$
\begin{array}{r}
x+3 \\
x^2-3x-10 \,\overline{\big)\, x^3+0x^2-13x+12} \\
-(x^3-3x^2-10x) \downarrow \\
\hline
3x^2-3x+12 \\
-(3x^2-9x-30) \\
\hline
6x+42 \text{ R}
\end{array}
$$

$\therefore$ O.A. @ $y = x+3$

$R(x)=0$

$6x+42=0$

$x=-7$

$\therefore$ crosses oblique asymptote @ $(-7,-4)$

Example 5: Determine if $f(x)=\dfrac{2x^3-x^2+3}{x^2}$ crosses the oblique asymptote. If so, determine the point of intersection.

$$
\begin{array}{r}
2x-1 \\
x^2 \,\overline{\big)\, 2x^3-x^2+0x+3} \\
-(2x^3) \downarrow \downarrow \downarrow \\
\hline
-x^2+0x+3 \\
-(-x^2) \downarrow \downarrow \\
\hline
3 \text{R}.
\end{array}
$$

$\therefore$ O.A. @ $y=2x-1$

$R(x)=0$

$3 \neq 0 \Rightarrow$ Does not cross the oblique asymptote.